NABASA AMOS

SOFTWARE ENGINEER · FULL-STACK · AI INTEGRATION

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PROFESSIONAL SUMMARY

Software Engineer specializing in full-stack development (React, Node.js, Python) and AI integration. Proven experience building and deploying applications for low-connectivity environments, achieving measurable impacts such as a 50% reduction in manual processes and a 60% increase in data accuracy. Passionate about building scalable, efficient, and impactful technology.

WORK EXPERIENCE

Full Stack Developer Intern | **John Vince Engineering** *June 2024 – July 2024*

- Engineered and deployed HostelEase, a full-stack booking platform, serving 500+ students and reducing manual registration efforts by 50%.
- Implemented secure JWT authentication, real-time availability filters, and a responsive UI to streamline the user experience from discovery to booking.

Software Developer Intern | **CAMTech Uganda** *September 2023 – October 2023*

- Developed a health-focused voicebot using Python and the ChatGPT API, enabling 100+ users in underserved areas to access critical health information via touchless voice queries.
- Led project management for the MVP, ensuring on-time delivery by coordinating a cross-functional team.

Site Verification & Reporting Engineer | **Hammer Uganda** *July 2025 – September 2025*

- Analyzed drive test log files from field measurements to generate SSV reports.
- Evaluated network performances across multiple generations.

Freelance Software Developer | **Self-Employed** *August 2025 – Present*

Xemora — Biometric Intelligence SaaS Platform *(React, TypeScript, Node.js, GraphQL, Python, FastAPI, PostgreSQL, pgvector, Gemini AI, Docker)*

- Built and deployed a production-grade SaaS platform combining Computer Vision, Biometric Security, and Generative AI (RAG) across a distributed microservices architecture, live at prism-vision.vercel.app.
- Implemented passwordless Face ID authentication using 512-dimensional FaceNet512 vector embeddings with real-time liveness and anti-spoofing detection via emotion micro-expression analysis.
- Engineered a RAG document intelligence module using Google Gemini 1.5 Flash embeddings and pgvector for hybrid context-aware vs. general-knowledge query routing, plus a voice command layer with real-time spectral biometric visualisation.
- Deployed across Vercel (React frontend), Render (Node.js GraphQL gateway), and Hugging Face Spaces (Dockerized Python AI engine); implemented a Round-Robin API key rotator to multiply free-tier throughput without service degradation.

AuthHub — Full-Stack Identity Provider *(Node.js, TypeScript, Express 5, PostgreSQL, Redis, React, Stripe)*

- Engineered a production-ready, multi-tenant Identity Provider with full OIDC/OAuth 2.0 compliance including Authorization Code Flow with PKCE, MFA via TOTP, Passkeys (WebAuthn), and dynamic RS256 JWT key rotation.
- Integrated Stripe for subscription billing, injecting plan entitlements directly into JWT scopes for downstream enforcement; built a self-service developer portal for tenants to register OAuth 2.0 clients and rotate secrets without admin intervention.

FaceSearch — Biometric Authentication System *(Python, FastAPI, Streamlit, FaceNet512, Docker)*

- Built a privacy-first, offline-capable facial recognition system using cosine similarity over FaceNet512 vector embeddings for 1:N face search, with real-time anti-spoofing liveness detection to prevent photo injection attacks.
- Exposed a FastAPI REST API for React/React Native integration alongside a Streamlit admin dashboard for enrollment management; containerised with Docker and automated CI/CD via GitHub Actions.

RentalTrack — Property Management Mobile App *(React Native, TypeScript, Expo, SQLite)*

- Developed an offline-first cross-platform mobile app for landlords covering tenant management, rent payment tracking with partial payment and credit balance support, and automated push notification reminders across multiple rent cycles.

- Shipped a production Android APK via Expo EAS Build (January 2026).

PaySpot — Captive Portal & Internet Access Management *(Node.js, JavaScript, TypeScript)*

- Built a web-based captive portal system that intercepts and gates internet access for users connected to managed networks, implementing session-based authentication and authorisation before granting full connectivity.

QuickSend — Multi-Channel Notification Platform *(Node.js, TypeScript, PostgreSQL, Redis, RabbitMQ)*

- Architecting a multi-tenant SaaS notification platform supporting email, SMS, push, and in-app delivery from a unified API, designed around event-driven messaging with Redis caching and RabbitMQ for reliable high-throughput dispatch.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++

Backend: Node.js, Express.js, Flask, REST APIs, GraphQL

Databases: MongoDB, MySQL, Firebase, PostgreSQL

Frontend: React, React Native, Electron, HTML5/CSS3, Tailwind CSS

AI / Data: TensorFlow, Pandas, NumPy, Scikit-learn

Tools & Platforms: Git, Docker, AWS (EC2, S3), Jenkins, Postman, Supabase

EDUCATION

Bachelor of Software Engineering *2022 – 2025*
Mbarara University of Science and Technology

Uganda Advanced Certificate of Education (A-Level) *2019 – 2020*
St. Mary's College Rushoroza

Uganda Certificate of Education (O-Level) *2015 – 2018*
Mbarara High School

Primary Leaving Examinations *2008 – 2014*
Kabale Preparatory School

LANGUAGES

English	Runyankore / Rukiga	Luganda	Spanish
Fluent	*Fluent*	*Good Proficiency*	*Basic*